

CouplingNet III: Projection and Green Reconstruction

Project in physical split variables, then reconstruct and average

STAGE 1: PHYSICAL BALANCE PROJECTION

COUPLINGNET RAW SPLIT

$$(\phi_{\text{raw}}, \psi_{\text{raw}})$$

not guaranteed balanced



PROJECTION STEP

$$r = f - (\phi_{\text{raw}} + \psi_{\text{raw}})$$
$$\phi = \phi_{\text{raw}} + \frac{1}{2}r, \quad \psi = \psi_{\text{raw}} + \frac{1}{2}r$$



BALANCED SPLIT

$$\phi + \psi = f$$

STAGE 2: GREEN RECONSTRUCTION

X-REPRESENTED SOLUTION

$$u_\phi = G_x[\phi]$$

Y-REPRESENTED SOLUTION

$$u_\psi = G_y[\psi]$$

FINAL AVERAGE

$$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$$

$$(\phi_{\text{raw}}, \psi_{\text{raw}}) \xrightarrow{\text{projection}} (\phi, \psi), \phi + \psi = f \xrightarrow{G_x, G_y} (u_\phi, u_\psi) \xrightarrow{\text{average}} u_{\text{pred}}.$$